

CS 498 QC ✧ Spring 2026

🌀 Homework 3 🌀

Due Tuesday, March 24th, 2026 at 9pm

- For this homework, you are encouraged to collaborate with your classmates. You should submit your solutions **individually**.
 - **Submit your solutions electronically on Gradescope as PDF files.**
 - Please format your solutions so that each problem begins on a new page and your name appears at the top of each page. While handwritten submissions are allowed, all homework submissions are **preferred to be typeset** using Latex or other equivalent programs.
 - Problems marked ★ are optional and usually a little harder.
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👉 Some important course policies 👈

- You are encouraged to collaborate with your classmates on homework problems. You are allowed to look at definitions of mathematical objects and general problem solving strategies as well. You should first make a serious attempt to solve the problems on your own. After that, you may use AI tools like ChatGPT, and you are encouraged to experiment with them.
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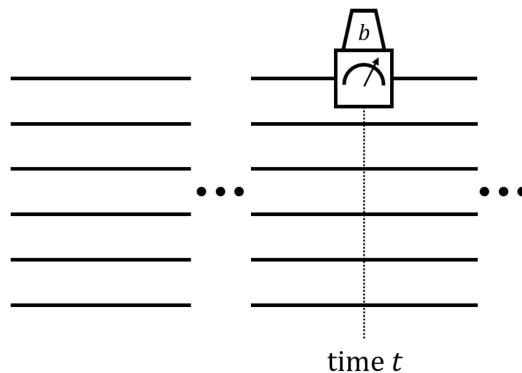
Note: The goal of these homework problems is to help you better understand quantum circuits and the fundamental quantum algorithms we have seen so far – Deutsch’s and Simon’s algorithm. In particular, you will work out: why the measurements in a quantum circuits can always be deferred until the end, a generalization of Deutsch’s algorithm and a modification of Simon’s problem (after some linear algebra refresher).

Problems:

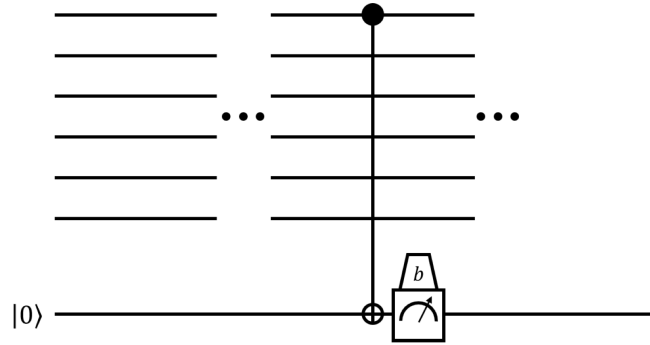
1. (15 points) Principle of deferred measurement.

In class, we mentioned without proof the fact that any quantum circuit containing intermediate (partial) measurements can be converted, with a small overhead, to one where all the measurements happen at the very end. This is often a convenient simplifying assumption when thinking about quantum circuits. In this problem, you will prove this fact.

Suppose we have an n -qubit quantum circuit, with some intermediate measurements, and we look at the first intermediate measurement gate that is applied. Suppose that it is applied to the first qubit at some time t . Let $|\psi\rangle$ denote the quantum state just prior to time t . When the measurement gate is applied, two things happen: first, one classical bit of information — call it b — appears on the measurement gate’s “readout”. Second, the state collapses according to the usual rules.



- (a) Suppose we do the following: First, we introduce a new $(n+1)$ -th qubit, initialized in the state $|0\rangle$. Second, we replace the measurement gate on the first qubit at time t with a CNOT gate whose control qubit is the first and whose target qubit is the $(n+1)$ -th. Finally, we immediately apply a measurement gate to the $(n+1)$ -th qubit, and treat its readout as “ b ”. Assume we then henceforth ignore the (collapsed) $(n+1)$ -th qubit. Show that the probability distribution over values of the bit b , and corresponding post-measurements states of the first n qubits is identical to the one obtained by the original circuit (i.e. by just measuring the first qubit without adding any auxiliary qubits). (*Hint: you may want to somehow write $|\psi\rangle = \alpha |0\rangle \otimes |\psi_0\rangle + \beta |1\rangle \otimes |\psi_1\rangle$.*)



- (b) Suppose that, even though the circuit we are running has intermediate measurements, it is *not* “adaptive”, meaning that its quantum gates do *not* depend on the outcomes of previous measurements. Then, consider the following circuit: instead of measuring the $(n + 1)$ -th qubit immediately after the CNOT gate, we defer its measurement to the very end. Deduce that this circuit gives rise to the same probability distribution on outcomes and post-measurement states (of the first n qubits) as the previous circuit (where the $(n + 1)$ -th qubit is measured immediately).

By repeating this for all intermediate measurement gates, we can always move all measurement gates to the end (at the cost of adding one extra qubit and one CNOT per deferral). We call this the “deferred measurement principle”.

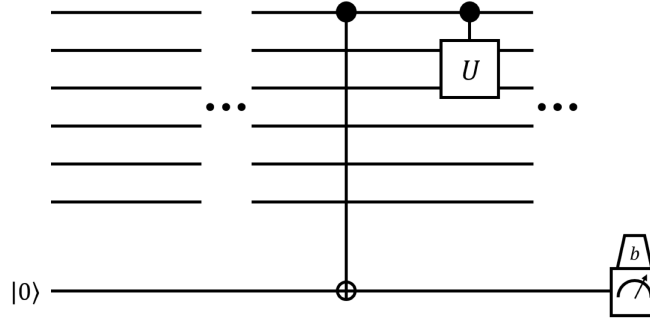
Suppose instead that the circuit you are running is “adaptive”, meaning that the rest of the circuit might include instructions of the form “apply the two-qubit gate U to qubits 2 and 3 if $b = 1$, else do nothing.” In particular, suppose that we don’t care about b ’s value per se; rather, we just want to do different subsequent quantum operations to the other qubits, depending on whether $b = 0$ or $b = 1$ (kind of like in the quantum teleportation protocol).

Given that, upon obtaining outcome b from measuring the first qubit, the post-measurement state of the first qubit is precisely $|b\rangle$, one can instead think of the above conditional instruction as a “controlled- U ” gate where qubit 1 is the “control” and qubits 2 and 3 are the “targets”, rather than as some interactive intervention wherein U is applied or not applied, depending on the outcome b . More formally, controlled- U is the linear transformation that acts as follows, for any two-qubit state $|\psi\rangle$:

$$|0\rangle |\psi\rangle \mapsto |0\rangle |\psi\rangle \quad \text{and} \quad |1\rangle |\psi\rangle \mapsto |1\rangle (U |\psi\rangle).$$

With this modification, the quantum circuit has exactly the same behaviour as before, but is no longer “adaptive”: the sequence of gates that is applied is fixed no matter what the measurement outcomes are (the only difference is that now some of these gates are controlled- U type of gates). Thus, the deferred measurement principle that

we showed in parts (a) and (b) for non-adaptive circuits can be applied to this modified circuit as well.



- (c) Show that controlled- U as defined above is a unitary transformation for any two-qubit unitary U .
- (d) Now, suppose we have compiled our circuit using “Controlled- U ” type of gates, and suppose that we have applied the deferred measurement principle. Since we don’t actually care to know the classical bit b , argue that in fact we don’t actually have to perform the measurement of the $(n + 1)$ -th qubit at all! (In other words, whether you measure the $(n + 1)$ -th qubit at the end or not does not affect the probability distribution of outcomes and corresponding post-measurement state of the first n qubits.)
- (e) At this point you may be wondering: do we really need to do the CNOT, either? Justify your answer either way.

2. (15 points) Generalizing Deutsch’s algorithm.

Let $n \in \mathbb{N}$. You are given black-box access to a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. You are *promised* that f is either *constant* (i.e. $f(x) = 0$ for all x or $f(x) = 1$ for all x) or *balanced* (i.e. $f(x) = 0$ on exactly half of the x ’s, and $f(x) = 1$ on the other half). The problem is to determine if f is constant or balanced.

- (a) How many queries to the black-box does a classical deterministic algorithm need to solve the problem (in the worst-case)?
- (b) It turns out that a classical *randomized* algorithm can do much better. Describe a classical randomized algorithm that makes at most 8 queries, and that satisfies the following: for all f (either constant or balanced), the algorithm guesses correctly, with probability at least 0.99, whether f is constant or balanced.
- (c) Let U_f be the $(n + 1)$ -qubit unitary that acts as follows: for all $x \in \{0, 1\}^n, y \in \{0, 1\}$,

$$U_f |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle .$$

Describe a circuit that makes one call to U_f and implements the following unitary O_f (possibly using auxiliary qubits that are returned to their initial state at the end of the circuit):

$$O_f |x\rangle = (-1)^{f(x)} |x\rangle .$$

- (d) Consider the following n -qubit circuit: start with $|0^n\rangle$, apply $H^{\otimes n}$, apply O_f , apply $H^{\otimes n}$, and finally measure all n qubits in the standard basis.

Show that by looking at the outcome of the measurement one can determine with certainty whether f is constant or balanced.

Linear Algebra Modulo 2 Refresher.

The integers modulo 2 constitute what's called a "field" in mathematics: a set of numbers (namely 0 and 1 here) for which all the standard operations of plus, minus, times, and division-by-nonzero work as expected. This "field" is sometimes denoted $\mathbb{F}_2$. (Other examples of fields include the real numbers, the complex numbers, the rational numbers, and the integers modulo p whenever p is prime.) It's a wonderful fact that pretty much all of linear algebra works just as well when the underlying scalars come from any fixed field, like $\mathbb{F}_2$. The set of all n -dimensional vectors in this case is denoted $\mathbb{F}_2^n$. You're most used to the cases $\mathbb{R}^n$ and $\mathbb{C}^n$, when the scalars are reals and complexes, respectively. Here we highlight some basic definitions and differences as preparation for the next problem:

- There's one pitfall to watch out for over finite fields: the "dot product" in $\mathbb{F}_2^n$, namely the operation $u \cdot v = \sum_{i=1}^n u_i v_i \pmod{2}$ (of course), doesn't act like an "inner product", in the sense that it's perfectly possible to have $u \cdot u = 0$ even though $u \neq 0$. Because of this, notions like "orthogonal basis" or " u and v are perpendicular" don't make as much sense in $\mathbb{F}_2^n$. We still frequently use the dot product operation anyway, though as in Simon's algorithm.
- A set of vectors $u_1, \dots, u_k$ is said to be linearly independent if the only linear combination $c_1 u_1 + \dots + c_k u_k$ that equals 0 is the trivial one with $c_1 = c_2 = \dots = c_k = 0$. As usual, the span (set of all linear combinations) of a set of k linearly independent vectors is called a k -dimensional subspace.
- As usual in linear algebra, one can study solutions of systems of linear equations like $Ax = 0$ (where $A \in \mathbb{F}_2^{m \times n}$ is a matrix and x is a vector of n unknowns) or, more generally, $Ax = b$ (where $b \in \mathbb{F}_2^n$ is a fixed right-hand side). The usual facts about Gaussian Elimination apply. The set of solutions x to $Ax = 0$ forms a subspace of dimension equal to $n - r$, where r is the maximum size of a linearly independent set of rows of A . Try to prove this yourself.

3. (20 points) Modified Simon's Algorithm.

In this problem you will analyze a variant of Simon's problem. Suppose you are given black-box query access to a function $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ that encodes **two** secrets $K, L \in \{0, 1\}^n$. This means that

$$f(x) = f(y) \quad \text{if and only if} \quad x + y \in \{0^n, K, L, K + L\}$$

We will assume that K and L are both nonzero and are not equal to each other.

Note: The sum $x + y$ of two bit strings is another string obtained by taking the bitwise XOR of x and y , or in other words, it is the addition of two vectors over the finite field $\mathbb{F}_2$.

- (a) Given such a function f , what is the size of its range? Meaning, how many possible strings can be output by f ?
- (b) Write a classical and quantum hybrid algorithm that makes $\text{poly}(n)$ queries to f , and with high probability outputs the secrets K and L . You should use Simon's algorithm for inspiration.
- (c) Prove that your algorithm works.
- (d) What if we now considered the situation where the function f is hiding not two but r secrets $L^{(1)}, \dots, L^{(r)}$ (which are all nonzero and distinct strings). How would your answers to parts (b) and (c) change?